Lab 4: Deciphering Classical Ciphers

In this lab you will decrypt (decipher) two messages that are encrypted using two of the classical ciphers. The first cipher is obtained by encrypting the plaintext using a shift cipher; the second with substitution cipher. In both cases, the plaintext is an English text.

For each cipher, provide detailed answers to the questions following the ciphers in this document and submit the document in Canvas.

A Shift Cipher: What does the text say?

Decipher this message by hand. You can resort to online tools for checking your work but your explanation on how you decipher it should be based on your manual work.

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fxxmjbfxltnslymwtzlmymjbttixxmjrjybnymfbtqkbmtmfifajwdlwjfyrnsiytjfymj
wzugzymjifwjistygjhfzxjtkxtrjbttihzyyjwxbtwpnslsjfwgdnsymjktwjxymjfxpj
imjwbmjwjxmjbfxltnslymjuttwhmnqibmtinistypstbymfynybfxifsljwtzxytxyfdf
siyfqpytfbtqkxfniytmnrnfrltnslytxjjrdlwfsirtymjwfsihfwwdmjwfhfpjfsifqn
yyqjutytkgzyyjwkwtrrdrtymjw
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1. The above text is obtained using the shift cipher.
    a. Which method would you suggest be used to break this cipher? Brute force or frequency analysis? Why?
    I recommend brute force because there are only 26 possible shifts, meaning you will be able to find the shift key in a MAXIMUM of 25 tries.
    b. Use the method you suggested in part a to break the cipher. Provide explanations for each step.
    I took the first 5 letters, shifted them over 1 in the alphabet, leaving me with "gyynk", which does not turn into any word. Then I tried a shift of 2, all the way through a shift of 21, which got me "asshe" or "as she" which are real words, then deciphered the rest.
    c. What is the plaintext?
    asshewasgoingthroughthewoodsshemetwithawolfwhohadaverygreatmindtoeatherupbuthedarednotbecauseofsomewoodcuttersworkingnearbyintheforestheaskedherwhereshewasgoingthepoorchildwhodidnotknowthatitwasdangeroustostayandtalktoawolfsaidtohimiamgoingtoseemygrandmotherandcarryheracakeandalittlepotofbutterfrommymother
    d. What is the key that was used to encrypt the message?
    21

A Substitution Cipher: What does the text say?

Decipher this message by hand. You can resort to the tools at the link below to automate certain parts of the process. In your explanations describe the whole process of how you ended up figuring out the key.

CJ IXO ZDXXHJS DYKA ZDJS OTX MDJHYAYJUC TYIIOJMC JZ KXW TESMCTYDX; IXO ZDXXHJS DYKA ZDJS OTX SYATOL SJUKOEYKC JZ KXW LJDR; IXO ZDXXHJS DYKA ZDJS OTX TXYATOXKYKA EIIXATXKYXC JZ MXKKCLIVEKYE; IXO ZDXXHJS DYKA ZDJS OTX CKJW-GEMMXH DJGRYXC JZ GJIJDEHJ; IXO ZDXXHJS DYKA ZDJS OTX GUDVEGXJUC CIJMXC JZ GEIYZJDKYE. QUO KJO JKIL OTEO. IXO ZDXXHJS DYKA ZDJS COJKX SJUKOEYK JZ AXJDAYE; IXO ZDXXHJS DYKA ZDJS IJJRJUO SJUKOEYK JZ OXKKXCCXX; IXO ZDXXHJS DYKA ZDJS XVXDL TYII EKH SJIX TYII JZ SYCCYCCYMMY. ZDJS XVXDL SJUKOEYKCYHX, IXO ZDXXHJS DYKA.

2. The above text is obtained using the substitution cipher.
    a. Which method would you suggest be used to break this cipher? Brute force or frequency analysis? Why?
    Frequency analysis is the best in this situation because there are 26^2 different combinations (I believe) that could be used for a substitution sypher, making it almost impossible to do that by hand in a timely manner.
    b. Use the method you suggested in part a to break the cipher. Provide explanations for each step.
    I used the links below to find the frequency of letters and patterns to make the connections between the letters.
    c. What is the plaintext?
    so let freedom ring from the prodigious hilltops of new hampshire; let freedom ring from the mighty mountains of new york; let freedom ring from the heightening alleghenies of pennsylvania; let freedom ring from the snow-capped rockies of colorado; let freedom ring from the curvaceous slopes of california. but not only that. let freedom ring from stone mountain of georgia; let freedom ring from lookout mountain of tennessee; let freedom ring from every hill and mole hill of mississippi. from every mountainside, let freedom ring.
    d. What is the key that was used to encrypt the message?
    eqghxzatynriskjmpdcouvwblf

You can use the tools at the links below for frequency analysis

http://practicalcryptography.com/cryptanalysis/text-characterisation/monogram-bigram-and-trigram-frequency-counts/

http://www.simonsingh.net/The_Black_Chamber/substitutioncrackingtool.html

You can also download and install CrypTool 1 for frequency analysis, which requires a Win32 environment:

https://www.cryptool.org/en/ct1/downloads/

Grading Rubric:

	MaxPoints
Deciphering Shift Cipher including answers to questions	4
Deciphering Substitution Cipher including answers to the questions	6
Total	10